CAPTURE THE BITSTREAM

Hackathon Problem Statements

Suggested Difficulty Ratings

1. Matrix Computation Accelerator – **10/10**
2. Signal Processing Engine – **9/10**
3. AXI-Based Scientific Calculator – **8/10**
4. Signal Generator – **4/10**
5. Even/Odd Signal Classifier – **3/10**

Problem 1: Reconfigurable 5×5 Matrix Computation Accelerator

Difficulty: 10/10

Objective: Design a hardware accelerator capable of performing different 5×5 matrix operations using Dynamic Function eXchange (DFX).

Reconfigurable Modules:
• RM1 – Matrix Addition
• RM2 – Matrix Multiplication
• RM3 – Matrix Transpose

Teams should accept two 5×5 integer matrices from a host PC, communicate via UART with a Python GUI, and display both inputs and outputs.

Problem 2: Reconfigurable Signal Processing Engine

Difficulty: 9/10

Objective: Build a runtime-reconfigurable DSP accelerator.

Reconfigurable Modules:
• RM1 – Moving Average Filter
• RM2 – FIR Low-Pass Filter
• RM3 – Peak Detection Engine

Signal samples are received from a host, processed by the active RM, and visualized using a Python GUI over UART.

Problem 3: Reconfigurable AXI-Based Scientific Calculator

Difficulty: 8/10

Objective: Design an AXI4-Lite hardware calculator on PYNQ with DFX.

Reconfigurable Modules:
• RM1 – Arithmetic (Add, Subtract, Multiply, Divide)
• RM2 – Integer Logic (AND, OR, XOR, Left Shift)
• RM3 – Decision Engine (Maximum, Minimum, Equality, Absolute Difference)

Only the compute engine is reconfigured while AXI infrastructure remains static.

Problem 4: Reconfigurable Signal Generator

Difficulty: 4/10

Objective: Generate different waveform types through DFX.

Reconfigurable Modules:
• RM1 – Square Wave Generator
• RM2 – Sawtooth/Ramp Generator
• RM3 – Impulse Train Generator

Output waveform samples through UART to a Python plotting GUI (or PWM to an LED).

Problem 5: Reconfigurable Even/Odd Signal Classifier

Difficulty: 3/10

Objective: Classify finite-length signals based on symmetry.

Reconfigurable Modules:
• RM1 – Even Signal Checker ($x[n] = x[-n]$)
• RM2 – Odd Signal Checker ($x[n] = -x[-n]$)

Sequences are stored in BRAM and results are indicated using LEDs.